Solve with us 8 - Not Graded

This assignment will not be graded and is only for practice.

Key Points: 1

- Two non-zero vectors u and v from an inner product space V are said to be orthogonal if $\langle u, v\rangle = 0$.
- A set S of non-zero vectors is said to be an orthogonal set of an inner product space if elements of the set S are mutually orthogonal. i.e., $\langle v_i, v_j\rangle = 0$ for $i \neq j$, any $v_i, v_j \in S$.
- Let $\{v_1, v_2, \ldots, v_n\}$ be an orthogonal set of vectors in an inner product space. Then $\{v_1, v_2, \ldots, v_n\}$ is a linearly independent set of vectors.
- Orthogonal basis: Let V be an inner product space. A basis consisting of mutually orthogonal vectors is called an orthogonal basis. In other word, a maximal orthogonal set is a basis and is called an orthogonal basis.
- An orthonormal set of vectors of an inner product space V is an orthogonal set of vectors such that the norm of each vector of the set is 1.
- An orthonormal basis is an orthonormal set of vectors which form a basis.
- Obtaining orthonormal set from orthogonal set: If $\gamma = \{v_1, v_2, \ldots, v_n\}$ is an orthogonal set of vectors then we can obtain an orthonormal set of vectors β from γ by dividing each vector by its norm i.e.,
$$\beta = \left\{\frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|}, \ldots, \frac{v_n}{\|v_n\|}\right\}.$$

Consider a set of vectors $S = \{(1, -1, -1), (1, 2, 0), (-1, 1, 1), (-1, \frac{1}{2}, -\frac{3}{2}), (0, 3, 1), (2, 1, 1), (0, 1, -1)\}$ from the inner product space $\mathbb{R}^3$ with usual inner product (i.e., the dot product).
Answer questions (1) and (2).

1 point
Which of the following subsets of the set S consist of mutually orthogonal vectors?

- [ ] $\{(1, -1, -1), (1, 2, 0), (-1, 1, 1)\}$
- [ ] $\{(1, -1, -1), (-1, \frac{1}{2}, -\frac{3}{2})\}$
- [ ] $\{(1, -1, -1), (-1, \frac{1}{2}, -\frac{3}{2}), (1, 2, 0)\}$
- [ ] S
- [ ] $\{(2, 1, 1), (1, -1, -1), (0, 1, -1)\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(1, -1, -1), (-1, \frac{1}{2}, -\frac{3}{2})\}$
$\{(2, 1, 1), (1, -1, -1), (0, 1, -1)\}$

1 point
Which of the following option(s) is (are) true?

☐ $\{(2,1,1),(0,1,-1)\}$ is an orthogonal basis of a vector space which is isomorphic to R^2.

☐ $\{(2,1,1)\}$ is not an orthogonal basis of a vector space which is isomorphic to R.

☐ $\{(1,-1,-1),(1,2,0)\}$ is an orthogonal basis of a vector space which is isomorphic to R^2.

☐ $\{(-1,1,1),(-1,\frac{1}{2},-\frac{3}{2}),(0,3,1),\}$ is an orthogonal basis of R^3.

☐ $\{\frac{(2,1,1)}{\sqrt{6}},\frac{(1,-1,-1)}{\sqrt{3}},\frac{(0,1,-1)}{\sqrt{2}}\}$ is an orthonormal basis of R^3.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\{(2,1,1),(0,1,-1)\}$ is an orthogonal basis of a vector space which is isomorphic to R^2.

$\{\frac{(2,1,1)}{\sqrt{6}},\frac{(1,-1,-1)}{\sqrt{3}},\frac{(0,1,-1)}{\sqrt{2}}\}$ is an orthonormal basis of R^3.

Key Points: 2

- The projection of a vector to a subspace: Let V be an inner product space, $v \in V$ and $W \subseteq V$ be a subspace. Then the projection of v onto W is the vector in W, denoted by $proj_W(v)$, computed as follows:

  Find an orthonormal basis $\{v_1, v_2, \ldots, v_n\}$ for W.

  Then $proj_W(v) = \sum_{i=1}^{n} \langle v, v_i \rangle v_i$.

  Remark: $proj_W(v)$ is independent of the chosen orthonormal basis.
- Let V be an inner product space and W be a subspace. Then the projection of vectors in V to W is a linear transformation from V to V with image W.

1 point

Consider the vector space R^3 with usual inner product. Which of the following vectors represents projection of the vector $u = (2,1,-2)$ onto the subspace whose an orthonormal basis is

$\{(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}), (\frac{-2}{3}, \frac{1}{3}, \frac{2}{3})\}$\{(2
√
1, 0, 2
√
1), (3−2, 31, 32)}?

◯
$(\frac{14}{9}, \frac{-7}{9}, \frac{-14}{9})$(914, 9−7, 9−14)

◯ (1,1,1)

◯ (2,1,-2)

◯
$(\sqrt{2}, 0, \sqrt{2})$(2
√
, 0, 2
√
)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(\frac{14}{9}, \frac{-7}{9}, \frac{-14}{9})$(914, 9−7, 9−14)

Consider the vector space R^3 R3 with usual inner product. If (a, b, c)(a, b, c) is the projection vector obtained from projecting the vector (1, 0, 1)(1, 0, 1) onto the subspace $\{(x, y, z) \mid x - y + z = 0, x, y, z \in R\}$\{(x, y, z) ∣ x − y + z = 0, x, y, z ∈ R\}, then find the value $3(a + b + c)$3(a + b + c).

Hint:Hint: Find any orthonormal basis of the subspace $\{(x, y, z) \mid x - y + z = 0, x, y, z \in R\}$ \{(x, y, z) ∣ x − y + z = 0, x, y, z ∈ R\} and use the definition of the projection of a vector onto a subspace.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 4
1 point

Key Points: 3Key Points: 3

- Gram-schmidt process:Gram-schmidt process: The Gram-schmidt process is used to get an orthogonal basis from any given basis of a vector space. A slight modification yields an orthonormal basis.
- Let $\gamma = \{v_1, v_2, \dots, v_n\}$γ = \{v1, v2, … , vn\} be a given ordered basis of a vector space. The steps in the process are described below and yield an orthonormal basis $\beta = \{u_1, u_2, \dots, u_n\}$β = \{u1, u2 , … , un\}.

Step-1 $w_1 = v_1$ w1 = v1 , and $u_1 = \frac{w_1}{\|w_1\|}$u1 = ‖w1‖w1.

Step-2 $w_2 = v_2 - \langle v_2, u_1 \rangle u_1$ w2 = v2 − ⟨v2, u1⟩u1 and $u_2 = \frac{w_2}{\|w_2\|}$u2 = ‖w2‖w2.

Step-3 $w_3 = v_3 - \langle v_3, u_1 \rangle u_1 - \langle v_3, u_2 \rangle u_2$ w3 = v3 − ⟨v3, u1⟩u1 − ⟨v3, u2⟩u2 and $u_3 = \frac{w_3}{\|w_3\|}$u3 = ‖w3‖w3.

⋮ ⋮

Step-i $w_i = v_i - \sum_{j=1}^{i-1} \langle v_i, u_j \rangle u_j$ and $u_i = \frac{w_i}{\|w_i\|}$.

⋮ ⋮

Step-n $w_n = v_n - \sum_{j=1}^{n-1} \langle v_n, u_j \rangle u_j$ and $u_n = \frac{w_n}{\|w_n\|}$.

Let β be an ordered basis of the inner product space $\mathbb{R}^3$ with usual inner product, where $\beta = \{(1,1,-1),(1,1,0),(1,0,0)\}$ and $\gamma = \{u_1, u_2, u_3\}$ be the corresponding orthonormal basis obtained using the Gram-schmidt process. Use this information to answer the questions (5), (6) and (7).

If $u_1 = (a_1, b_1, c_1)$, then find the value of $\sqrt{3}(a_1 + b_1 + c_1)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

If $u_2 = (a_2, b_2, c_2)$, then find the value of $\sqrt{6}(a_2 + b_2 + c_2)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 4
1 point

If $u_3 = (a_3, b_3, c_3)$, then find the value of $\frac{1}{\sqrt{2}}(a_3 + b_3 + c_3)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point

Key Points: 6

- Orthogonal transformation: Let V be an inner product space and T be a linear transformation from V to V. T is said to be an orthogonal transformation if

$$\langle Tv, Tw \rangle = \langle v, w \rangle, \ \forall v, w \in V$$

- Orthogonal matrix: A square matrix A is called an orthogonal matrix if $AA^T = A^TA = I$ (identity matrix)

1 point

Consider the following linear transformations $T_1, T_2 : \mathbb{R}^2 \to \mathbb{R}^2$, and $T_3, T_4 : \mathbb{R}^3 \to \mathbb{R}^2$ defined as

$$T_1(x, y) = \left(\frac{x+y}{\sqrt{2}}, \frac{x-y}{\sqrt{2}}\right),$$

$$T_2(x, y) = (x + 3y, 2x - 2y),$$

$$T_3(x, y, z) = \left(\frac{x+z}{\sqrt{2}}, \frac{x+y\sqrt{3}+z}{\sqrt{3}}\right),$$

$$T_4(x, y, z) = (x + y + z, x + y),$$

where R^3 and R^2 are the inner product spaces with respect to the dot product. Which of the linear transformation is an orthogonal transformation?

- ○ T_1
- ○ T_2
- ○ T_3
- ○ T_4

No, the answer is incorrect.
Score: 0
Accepted Answers:
T_1

1 point

Which of the following option(s) is (are) true?

- ☐ $\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$ is an orthogonal matrix.
- ☐ $\frac{1}{\sqrt{2}}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}$ is an orthogonal matrix.
- ☐

$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 2 \\ -1 & -2 & 1 \end{bmatrix}$ 12−101−2121 is an orthogonal matrix

☐

$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix}$ 2

√

1 1010 2

√

010−1 is an orthogonal matrix.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$[100−1] is an orthogonal matrix.

$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ 2

√

1 [11−11] is an orthogonal matrix.

$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix}$ 2

√

1 1010 2

√

010−1 is an orthogonal matrix.